Applied Machine Learning Project Report

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Machine Learning–Driven Formula 1 Race Strategy Dashboard

1. Introduction

Formula 1 race strategy involves complex, time-sensitive decision-making under uncertainty, including when to pit, how tyre degradation affects lap times, and whether overtaking opportunities are likely to arise. Traditional dashboards provide descriptive statistics but lack predictive capability.

This project presents an applied machine learning decision-support system for Formula 1 race strategy. Using historical race telemetry from the 2023 Italian Grand Prix (Monza), multiple machine learning models are trained and deployed within a real-time interactive dashboard to support strategic decisions during a race.

The objective of this project is not merely to visualise data, but to apply machine learning models to predict future outcomes such as lap times, pit stop timing, overtaking probability, and post-pit position.

2. Dataset and Preprocessing

Data Source

The dataset was obtained using the FastF1 Python library, which provides official Formula 1 timing, lap, and telemetry data. The race selected for this study was the 2023 Italian Grand Prix (Monza), chosen due to its strategic sensitivity to tyre degradation and pit timing.

Data Cleaning

Only competitive laps were retained using FastF1's `pick_quicklaps()` function. Laps with missing lap time values were removed. Lap times were converted to seconds for numerical modelling.

Categorical variables such as Driver and Tyre compound were encoded using Label Encoding, enabling their use in tree-based machine learning models.

3. Features

Key engineered features include:

- Tyre Age: Number of laps completed since the start of the current stint.
- Encoded Driver ID: Captures driver-specific performance characteristics.
- Encoded Tyre Compound: Differentiates between soft, medium, and hard compounds.
- Position Change: Used to label overtaking events.
- Next Position: Used as a proxy for post-pit rejoin position.

4. Machine Learning Models

Lap Time Prediction (Regression)

A Random Forest Regressor was trained to predict future lap times based on driver, tyre compound, and tyre age.

- Input features: Driver, compound, tyre age
- Output: Predicted lap time (seconds)

Pit Stop Decision Model (Classification)

A Random Forest Classifier predicts whether a driver should Stay Out, Pit Soon, or Pit Now. Labels were generated using weak supervision based on tyre age thresholds.

Overtake Probability Model (Classification)

Predicts the probability of gaining a position in the next race phase. Target variable is a binary indicator of position improvement.

Post-Pit Position Prediction (Regression)

Predicts the expected position after a pit stop, answering the key strategic question: Where will this driver rejoin if they pit now?

Driver	Pos	Tyre	Tyre Age	Pred Lap	Pit ML	Overtake %	Pred Pos After Pit
VER	1	HARD	30	87.82	PIT_SOON	0.0%	1
PER	2	HARD	29	86.08	PIT_SOON	0.0%	2
SAI	3	HARD	31	86.95	PIT_SOON	0.0%	4
LEC	4	HARD	30	86.83	PIT_SOON	0.0%	6
RUS	5	HARD	31	86.69	PIT_SOON	0.0%	6
HAM	6	MEDIUM	23	86.07	PIT_NOW	2.0%	5
ALB	7	HARD	35	87.32	PIT_SOON	0.0%	8
NOR	8	HARD	28	87.33	PIT_SOON	0.0%	8
ALO	9	HARD	29	87.31	PIT_SOON	0.0%	9
BOT	10	MEDIUM	25	86.97	PIT_NOW	0.0%	11
PIA	11	MEDIUM	51	85.73	PIT_NOW	95.3%	11
LAW	12	MEDIUM	50	86.28	PIT_NOW	78.0%	12
SAR	13	HARD	36	87.49	PIT_SOON	0.0%	12
ZHO	14	MEDIUM	50	86.85	PIT_NOW	82.0%	14
GAS	15	MEDIUM	50	87.07	PIT_NOW	76.0%	15
OCO	15	HARD	26	88.39	STAY_OUT	0.7%	16
STR	16	HARD	30	88.93	PIT_SOON	0.0%	15
HUL	17	MEDIUM	50	85.96	PIT_NOW	39.3%	17
MAG	18	MEDIUM	38	86.71	PIT_NOW	32.0%	15

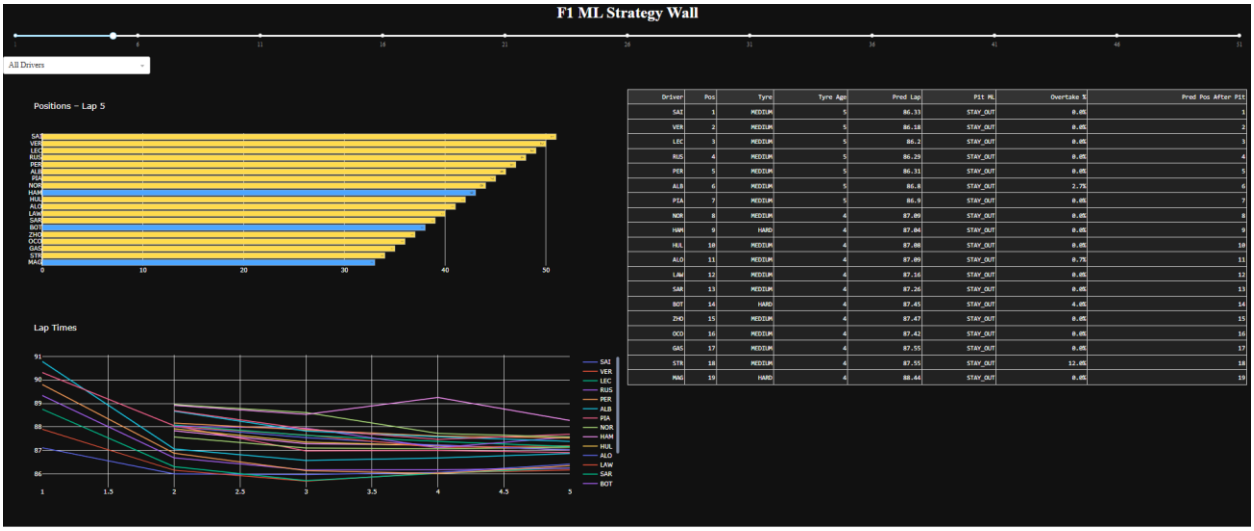
5. System Architecture

The system separates offline training of machine learning models from real-time inference within an interactive Dash application. A lap slider and automatic lap increment simulate real-time race progression, while models dynamically update predictions as tyre age and race context change.

6. Results and Dashboard Analysis

Early Stint Behaviour (Figure 1 – Lap 10)

At Lap 4, most drivers are on medium tyres with low tyre age. The pit decision model consistently outputs STAY_OUT. Overtake probabilities are low, and predicted post-pit positions closely match current positions.



Driver	Pos	Tyre	Tyre Age	Pred Lap	Pit ML	Overtake %	Pred Pos After Pit
SAI	1	MEDIUM	5	86.33	STAY_OUT	0.0%	1
VER	2	MEDIUM	5	86.18	STAY_OUT	0.0%	2
LEC	3	MEDIUM	5	86.2	STAY_OUT	0.0%	3
RUS	4	MEDIUM	5	86.29	STAY_OUT	0.0%	4
PER	5	MEDIUM	5	86.31	STAY_OUT	0.0%	5
ALB	6	MEDIUM	5	86.8	STAY_OUT	2.7%	6
PIA	7	MEDIUM	5	86.9	STAY_OUT	0.0%	7
NOR	8	MEDIUM	4	87.09	STAY_OUT	0.0%	8
HAM	9	HARD	4	87.04	STAY_OUT	0.0%	9
HUL	10	MEDIUM	4	87.08	STAY_OUT	0.0%	10
ALO	11	MEDIUM	4	87.09	STAY_OUT	0.7%	11
LAW	12	MEDIUM	4	87.16	STAY_OUT	0.0%	12
SAR	13	MEDIUM	4	87.26	STAY_OUT	0.0%	13
BOT	14	HARD	4	87.45	STAY_OUT	4.0%	14
ZHO	15	MEDIUM	4	87.47	STAY_OUT	0.0%	15
OCO	16	MEDIUM	4	87.42	STAY_OUT	0.0%	16
GAS	17	MEDIUM	4	87.55	STAY_OUT	0.0%	17
STR	18	MEDIUM	4	87.55	STAY_OUT	12.0%	18
MAG	19	HARD	4	88.44	STAY_OUT	0.0%	19

Mid-Stint Degradation and Strategic Shift

By Lap 20, tyre age increases significantly. The pit decision model predicts PIT_SOON for leading drivers. Overtake probabilities increase for midfield drivers, and predicted lap times show a gradual upward trend.

Driver	Pos	Tyre	Tyre Age	Pred Lap	Pit ML	Overtake %	Pred Pos After Pit
VER	1	HARD	30	87.02	PIT_SOON	0.0%	1
PER	2	HARD	29	86.00	PIT_SOON	0.0%	2
SAI	3	HARD	31	86.95	PIT_SOON	0.0%	3
LEC	4	HARD	30	86.83	PIT_SOON	0.0%	4
RUS	5	HARD	31	86.69	PIT_SOON	0.0%	5
HAM	6	MEDIUM	23	86.87	PIT_NOW	2.0%	6
ALB	7	HARD	35	87.12	PIT_SOON	0.0%	7
NOR	8	HARD	28	87.13	PIT_SOON	0.0%	8
ALO	9	HARD	29	87.31	PIT_SOON	0.0%	9
BOT	10	MEDIUM	25	86.97	PIT_NOW	0.0%	10
PIA	11	MEDIUM	51	85.73	PIT_NOW	95.3%	11
LAW	12	MEDIUM	50	86.28	PIT_NOW	78.0%	12
SAR	13	HARD	36	87.49	PIT_SOON	0.0%	13
ZHO	14	MEDIUM	50	86.85	PIT_NOW	82.0%	14
GAS	15	MEDIUM	50	87.07	PIT_NOW	76.0%	15
OCO	16	HARD	26	88.10	STAY_OUT	0.7%	16
STR	16	HARD	30	88.93	PIT_SOON	0.0%	16
HUL	17	MEDIUM	50	85.96	PIT_NOW	19.3%	17
MAG	18	MEDIUM	38	86.71	PIT_NOW	12.0%	18

7. Limitations

Limitations include:

- External factors like safety cars, traffic, and weather were not included.

8. Conclusion and Future Work

The project demonstrates the successful application of machine learning to Formula 1 race strategy, moving beyond descriptive analytics toward predictive and prescriptive decision support.